

Assignment 1/Module 2

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Assignment

1. If $R = 50 \, \Omega$, $L = 1.5 \, \text{H}$, what value of C will make an RLC series circuit: (a) overdamped, (b) critically damped, (c) underdamped?

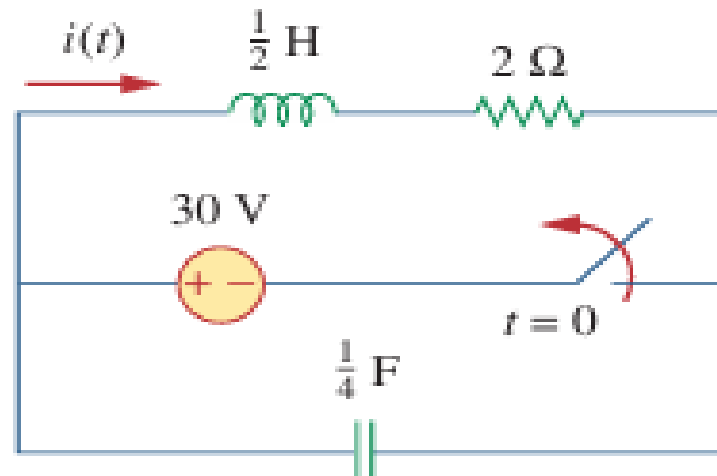
2. The responses of a series RLC circuit are

$$v_C(t) = 30 - 10e^{-20t} + 30e^{-10t} \, \text{V}$$

$$i_L(t) = 40e^{-20t} - 60e^{-10t} \, \text{mA}$$

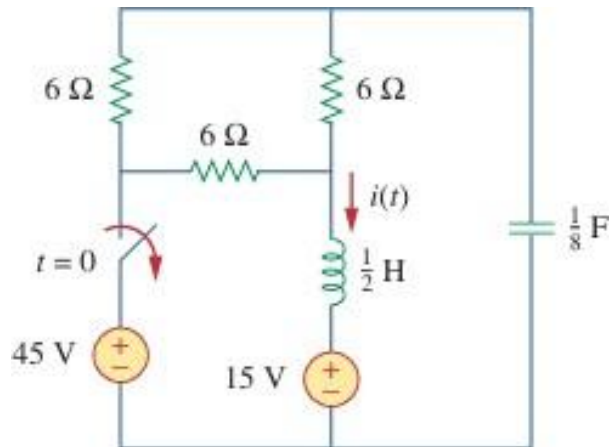
where v_C and i_L are the capacitor voltage and inductor current, respectively. Determine the values of R , L , and C .

3. The switch in the circuit has been closed for a long time but is opened at $t = 0$. Determine for $i(t)$ for $t > 0$.

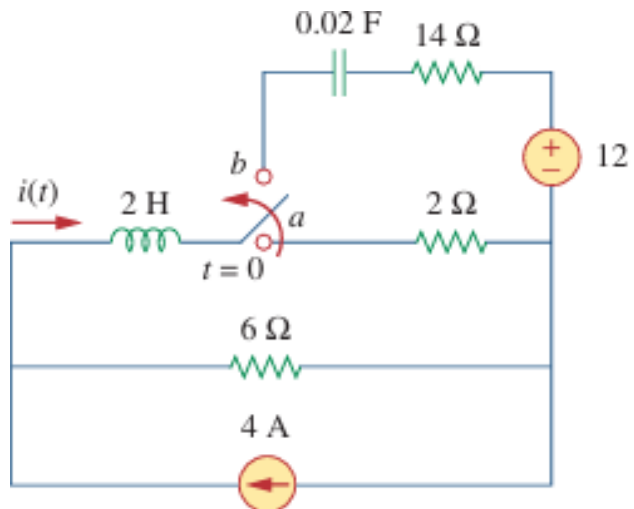


Assignment

4. For the network shown below, solve for $i(t)$ for $t > 0$.

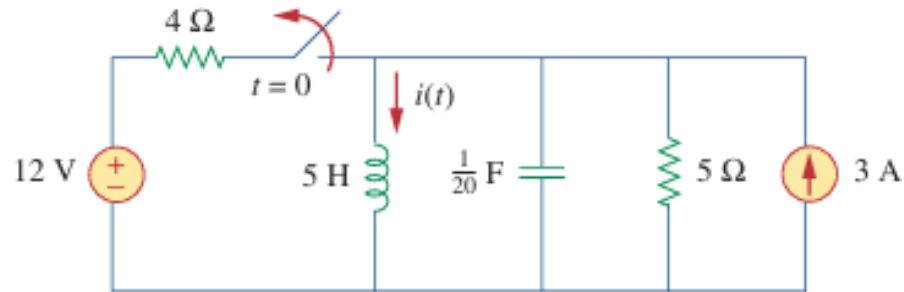


5. The switch in the circuit is moved from position a to b at $t = 0$. Determine $i(t)$ for $t > 0$.



Tutorial

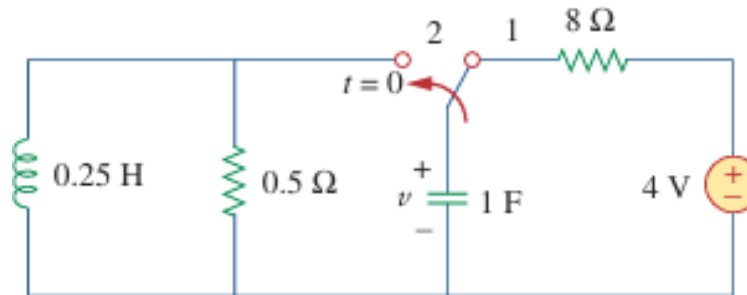
6. Determine $i(t)$ for $t > 0$ in the circuit.



7. In the circuit, the switch has been in position 1 for a long time but moved to position 2 at $t = 0$. Find:

(a) $v(0^+)$, $dv(0^+)/dt$,

(b) $v(t)$ for $t \geq 0$.



8. Find $v(t)$ for $t > 0$ in the circuit.

